

练习十三 统计物理学基础（一）

1. 相同；不同；相同；

2. 1:1 2:1 10:3

3. [2]

4. [2]

5. 由 $pV = \frac{m}{M}RT \Rightarrow M = \frac{RT\rho}{P} = 2 \times 10^{-3}$ 千克/摩尔 = 2 克/摩尔

$\therefore$ 该气体为氢气, $\sqrt{v^2} = \sqrt{\frac{3RT}{M}} = 1.93 \times 10^3 \text{ m/s}$

6. (1) $n = \frac{P}{kT} = \frac{1.013 \times 10^5}{1.38 \times 10^{-23} \times 400} = 1.8 \times 10^{25} \text{ m}^{-3}$

(2) $\mu = \frac{M}{N_0} = \frac{32 \times 10^{-3}}{6.02 \times 10^{23}} = 5.3 \times 10^{-26} \text{ kg}$

(3) $\rho = \frac{m}{V} = \frac{Mp}{RT} = 0.98 \text{ kg/m}^3$

(4) $E_k = n \times \frac{5}{2} kT = 2.5 \times 10^5 \text{ J}$